

M Safwan Hasan Khan

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ABOUT ME

4th-semester Software Engineering student at IUT with a focus on embedded systems and autonomous architectures. Proven experience in hardware-software integration through competitive robotics (SoccerBot) and real-time control systems. Dedicated to building robust, low-latency autonomous solutions for mission-critical applications.

EDUCATION

Islamic University of Technology (IUT)

Bachelor of Science in Software Engineering

Sept. 2024 – Present

Gazipur, Bangladesh

– **Relevant Coursework:** Data Structures, DBMS, OOP, Digital Logic Design, Algorithms, Linear Algebra.

Notre Dame College

Higher Secondary Certificate (HSC) | 100% Attendance

July 2022 – July 2024

Dhaka, Bangladesh

TECHNICAL SKILLS

Languages: C, C++, Java, Python, Dart (Flutter), JavaScript, TypeScript, SQL (PostgreSQL)

Tech Stacks: React, Node.js, Express.js, Spring Boot, Next.js, MERN Stack

Embedded & Tools: Arduino IDE, ESP32, Bluepad32, PWM, Git, VS Code

Other: Real-Time Motor Control, Hardware-Software Integration, System Architecture

PROJECTS

SoccerBot Motor Control | *ESP32, C++, PWM* | 🌐 GitHub

- Engineered a low-latency wireless control system using ESP32 and Bluepad32 for PS5 DualSense integration; implemented differential steering and throttle-mixing for 4-motor autonomous robot maneuvering.
- Designed RPM limiting and PWM constraint handling for performance optimization, foundational for drone motor speed controllers (ESCs) and real-time flight stability logic.

FeatherDB | *C++17, SQL Engine, RBAC* | 🌐 GitHub

- Architected a C++17 SQL engine with a custom tokenizer and AST-parser. Engineered a storage manager optimized for telemetry-grade data pipelines and persistent high-frequency logging.

RS_24_Orientation | *Arduino, Python, Serial* | 🌐 GitHub

- Built a real-time hardware-software bridge with deterministic state machines for millisecond-precision I/O modulation; demonstrated for a fun game for freshers in their orientation for IUTRS (Robotics Society).

3D Rendering Engine | *Java, Linear Algebra* | 🌐 GitHub

- Implemented 3D spatial coordinate transformations and projections foundational for SLAM and obstacle avoidance, simulating coordinate systems used in autonomous 3D flight environments.

ACHIEVEMENTS

1st Runner Up, SoccerBot Competition – Techathon 2.0 (Intra-IUT)

Aug. 2025

- Optimized hardware-software integration for real-time maneuverability and sensor-motor feedback.

2nd Runner Up, Voyagers Quiz – IUT Al-Fazari InterStellar Society

- Demonstrated rapid analytical problem-solving in a competitive STEM orientation context.

Hackathon Competitor, CodeRush 2025 | 🌐 GitHub | 📺 Demo

- Developed innovative solutions within tight deadlines; recognized for technical adaptability and performance.